4,212,1 Macroeconomics III
Exercises and Independent Studies

Maria Bolboaca

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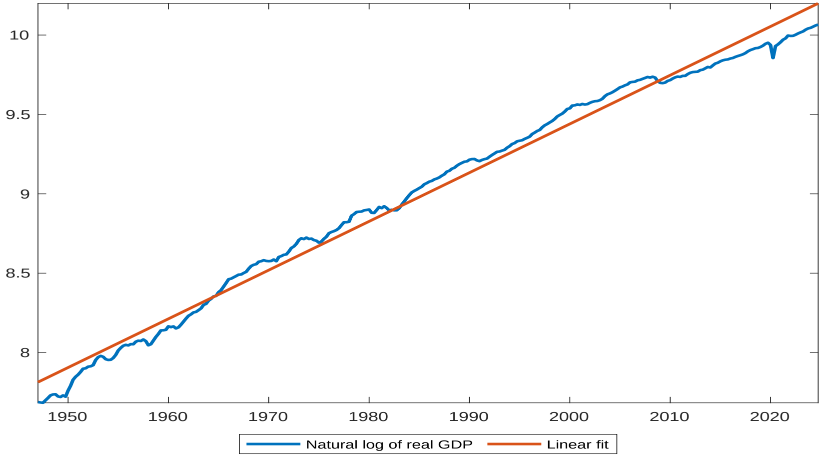
Outline

- Short Recap of Main Topics in Lecture 1
- Discussion of Results of Take-Home Exercises for Session 1

Long Run Growth

- Robert Lucas (Nobel Prize winner) said "Once you start to think about growth, it is difficult to think about anything else."



Technicalities (Recap)

- Why plot series in natural logs? GDP grows exponentially over time, implying the slope gets steeper as time passes. The log of an exponential function is a linear function, which is much easier to interpret. The slope of a plot in the log is approximately just the growth rate of the series.
- Why use real GDP instead of nominal GDP?
- GDP may change over time because prices change or quantities change. Changes in quantities are what we care about for welfare. To address this, real GDP uses constant prices over time to measure changes in output.
- Why real GDP per worker?
- GDP can go up either because more people are working in an economy or because the people working in an economy are producing more. For thinking about an economy's productive capacity and for making comparisons across time, we want a measure of GDP that controls for the number of people working in an economy.

Long Run Growth: Stylized Facts

- “Stylized” means that these facts are roughly true over sufficiently large periods.
- **Stylized fact 1:** Some countries are rich, and some are poor. However, there is some tendency towards a more equal world income distribution, though not much at the very bottom.
- **Stylized fact 2:** Growth rates vary substantially between countries, and by the process of growing or declining fast, a country can move from being relatively poor to being relatively rich, or from being relatively rich to being relatively poor.
- **Stylized fact 3:** Growth can break in a country, turning from a high rate to a low one or vice versa.
- **Stylized fact 4:** Convergence: If one controls appropriately for structural differences between the countries of the world, a lower initial value of GDP per worker tends to be associated with a higher subsequent growth rate in GDP per worker.

Long Run Growth: Stylized Facts

- **Stylized fact 5:** Real GDP per worker grows at a sustained, roughly constant, rate over long periods (in advanced economies).
- **Stylized fact 6:** Labor's share of GDP has stayed relatively constant, hence the average real wage of a worker has grown by approximately the same rate as GDP per worker.

$$Labourshare = \frac{w_t L_t}{Y_t}$$

- **Stylized fact 7:** Capital's share and the rate of return on capital have shown no trend, therefore the capital-output ratio K/Y has been relatively constant, and the capital intensity K/L has grown by approximately the same rate as GDP per worker.

$$Capitalshare = 1 - Labourshare = \frac{r_t K_t}{Y_t}$$

Balanced Growth

- GDP per worker, consumption per worker, the real wage, and the capital intensity all grow at the same constant rate, g.
- The labour force (population) grows at a constant rate, n.
- GDP, consumption, and capital grow at the common rate, g+n.
- The capital-output ratio and the rate of return on capital are constant.

What is a (Macro) Model?

- To an economist, a model is a simplified representation of the economy in which only the main ingredients are accounted for.
- Ideally, we would like to perform experiments like in other sciences, but this is possible only in a few areas of economics (behavioral economics). In macroeconomics, however, things are different.
- We build macro models to conduct experiments that we cannot run in the real world and use the results from these experiments to inform policy-makers.
- A macro (economic) model is composed of a set of mathematical relationships through which one determines how variables affect each other (direction of relationships) and quantifies their impacts.

The Basic Solow Model

- The Solow Model is the principal model for understanding long-run growth and cross-country income differences.
- Robert Solow won a Nobel Prize for his work!
- The key equations of the model are:
  1. an aggregate production function;
  2. a consumption/saving function;
  3. an accumulation equation for physical capital.

The Aggregate Production Function

- Denote K_t as the stock of capital and L_t as the total number of workers in period t.
- Let Y_t denote output produced in period t. Suppose that there is a single, representative firm that leases labor and capital from a single, representative household each period to produce output. The production function is given by:

$$Y_t = F(K_t, L_t).$$

The Aggregate Production Function

The function $F(\cdot)$ is assumed to have the following properties:

- $F'_K > 0$ and $F'_L > 0$ (i.e., marginal products are always positive: more of either input means more output);
- $F''_{KK} < 0$, $F''_{LL} < 0$ (i.e., diminishing marginal products in both factors: more of one factor means more output, but the more of the factor you have, the less an additional unit of that factor adds to output);
- $F''_{KL} > 0$ (i.e., if you have more capital, the marginal product of labor is higher);
- $F(\lambda K_t, \lambda L_t) = \lambda F(K_t, L_t)$ (i.e. production function has constant returns to scale);
- $F(0, L_t) = F(K_t, 0) = 0$ (i.e., both capital and labor are necessary to produce).

Problem of the Firm

The optimization problem of the firm is to choose capital and labor to maximize profits (Π_t):

$$\max_{K_t, L_t} \Pi_t = F(K_t, L_t) - w_t L_t - r_t K_t,$$

- where w_t denotes the real wage paid to workers and r_t denotes the real return to capital.
- The first order conditions (FOCs) are:

$$w_t = F'_L$$

$$r_t = F'_K$$

- FOCs imply that the firm ought to hire capital and labor to the point at which the marginal benefit of doing so (the marginal product of capital or labor) equals the marginal cost of doing so (the factor price).

Problem of the Household

- There exists a representative household in the economy. Its budget constraint is:

$$C_t + S_t = w_t L_t + r_t K_t + \Pi_t.$$

- Although separate decision-making units, the household owns the firm, hence dividend payment equals any profit earned by the firm.
- However, the firm earns no profit under the assumption of perfect competition ($w_t L_t + r_t K_t = Y_t$). In other words:

$$C_t + S_t = Y_t.$$

- "Biology":

$$L_{t+1} = (1 + n) L_t.$$

Capital Accumulation Equation

- The Solow model assumes that savings represent a constant fraction of output:

$$S_t = sY_t,$$

  where $0 < s < 1$ denotes the saving rate (equivalently, the investment rate $S_t = I_t$).

- The initial level of capital, K_t, is given, but future levels of capital can be influenced through investment.

$$K_{t+1} = sY_t + (1 - \delta)K_t,$$

  where $0 < \delta < 1$ is the depreciation rate (the fraction of the capital stock that becomes obsolete each period).

- Note that the real interest rate is defined as:

$$\rho_t = r_t - \delta.$$

The Complete Set of Equations

Assume: $F(K_t, L_t) = B(K_t)^{\alpha} L_t^{1-\alpha}$. Then:

$$
\begin{aligned}
Y_t &= B(K_t)^{\alpha} L_t^{1-\alpha}, && (1)\\
K_{t+1} &= S_t + (1-\delta)K_t, && (2)\\
w_t &= (1-\alpha)B(K_t)^{\alpha} L_t^{-\alpha}, && (3)\\
r_t &= \alpha B(K_t)^{\alpha-1} L_t^{1-\alpha}, && (4)\\
C_t + S_t &= Y_t, && (5)\\
S_t &= sY_t, && (6)\\
L_{t+1} &= (1+n)L_t, && (7)\\
\rho_t &= r_t - \delta. && (8)
\end{aligned}
$$

5 Parameters: $B, \alpha, s, n, and \delta$. 8 Endogenous variables: Y, K, L, r, w, S, C, and ρ of which K_t and L_t are state (predetermined) variables.

Transformations in per Worker Terms

$$\frac{Y_t}{L_t} = B\left(\frac{K_t}{L_t}\right)^{\alpha}$$

$$\frac{K_{t+1}}{L_{t+1}}\frac{L_{t+1}}{L_t} = s\frac{Y_t}{L_t} + (1-\delta)\frac{K_t}{L_t}$$

$$w_t = (1-\alpha)B\left(\frac{K_t}{L_t}\right)^{\alpha}$$

$$r_t = \alpha B\left(\frac{K_t}{L_t}\right)^{\alpha-1}$$

$$\frac{C_t}{L_t} + \frac{S_t}{L_t} = \frac{Y_t}{L_t}$$

$$\frac{S_t}{L_t} = s\frac{Y_t}{L_t}$$

$$\rho_t = r_t - \delta$$

Transformations in per Worker Terms

Define: $k_t = \frac{K_t}{L_t}$ and $y_t = \frac{Y_t}{L_t}$. Then:

$$\frac{K_{t+1}}{L_{t+1}}\frac{L_{t+1}}{L_t} = s\frac{Y_t}{L_t} + (1-\delta)\frac{K_t}{L_t},$$

and using that

$$L_{t+1} = (1+n)L_t,$$

we get

$$k_{t+1} = s\frac{y_t}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)},$$

Transformations in per Worker Terms

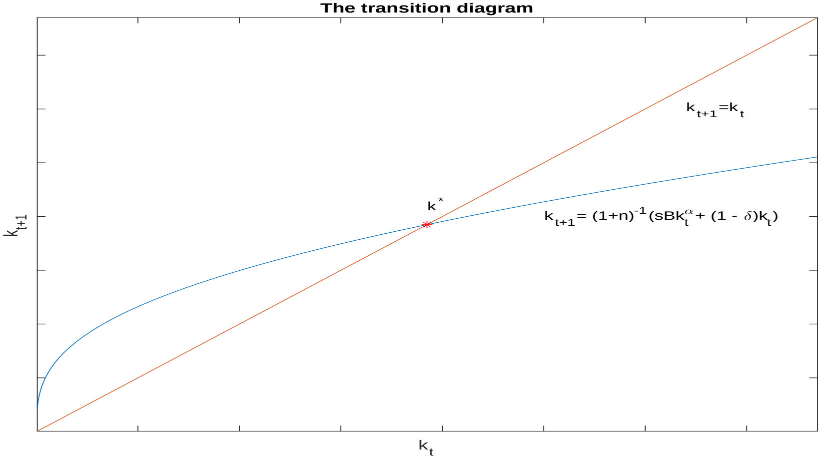
The set of equations with transformed variables in per-worker terms is:

$$
\begin{aligned}
y_t &= B(k_t)^{\alpha}, \\
k_{t+1} &= s\frac{y_t}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)}, \\
w_t &= (1-\alpha)B(k_t)^{\alpha}, \\
r_t &= \alpha B(k_t)^{\alpha-1}, \\
c_t + s_t &= y_t, \\
s_t &= s y_t, \\
\rho_t &= r_t - \delta.
\end{aligned}
$$

The Transition Equation

Replacing the production function: $y_t = B(k_t)^\alpha$ in the capital accumulation equation, we obtain the transition equation:

$$k_{t+1} = s\frac{Bk_t^\alpha}{(1+n)} + (1-\delta)\frac{k_t}{(1+n)}. \tag{9}$$



Slope of the Transition Equation

$$\frac{dk_{t+1}}{dk_t} = \frac{s\alpha B}{(1+n)} \frac{1}{k_t^{1-\alpha}} + \frac{(1-\delta)}{(1+n)}.$$

- If $k_t \to 0$ then the slope $\frac{dk_{t+1}}{dk_t} \to \infty$
- If $k_t \to \infty$ then the slope $\frac{dk_{t+1}}{dk_t} \to \frac{(1-\delta)}{(1+n)}$
- Interpretation: The plot starts out steep at the origin, and flattens out as k_t gets bigger, eventually having a slope equal to $\frac{(1-\delta)}{(1+n)} < 1$. The 45-degree line has a slope of 1. The k_{t+1} plot starts out with a slope greater than 1, but then it goes below 1. Since it is a continuous curve, this means that the plot of k_{t+1} crosses the 45-degree line exactly once away from the origin. We indicate this point with k^* and refer to it as the "steady state."

Steady State

- The steady state is a natural point of interest. This is not because the economy is always at the steady state, but rather because, no matter where the economy starts (provided it does not start with $k_t = 0$), it will naturally gravitate towards this point.
- To algebraically solve for the steady state capital stock, take the transition equation (eq. 9) and set $k_{t+1} = k_t = k^*$:

$$k^* = B^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{1}{(1-\alpha)}}$$

Steady State

$$k^* = B^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{1}{(1-\alpha)}},$$

$$y^* = B(k^*)^\alpha = B^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

$$w^* = (1-\alpha)B(k^*)^\alpha = (1-\alpha)B^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

$$r^* = \alpha B(k^*)^{\alpha-1} = \alpha \left(\frac{s}{n+\delta} \right)^{-1},$$

$$c^* = (1-s)y^* = (1-s)B^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

$$s^* = sy^* = sB^{\frac{1}{(1-\alpha)}} \left(\frac{s}{n+\delta} \right)^{\frac{\alpha}{(1-\alpha)}},$$

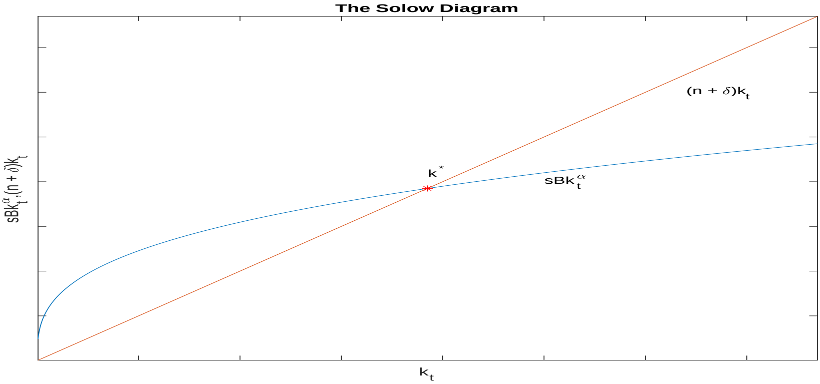
$$\rho^* = r^* - \delta = \alpha \left(\frac{s}{n+\delta} \right)^{-1} - \delta.$$

The Solow Equation

$$\Delta k_{t+1} = k_{t+1} - k_t = s\frac{Bk_t^{\alpha}}{(1+n)} - (n+\delta)\frac{k_t}{(1+n)}.$$

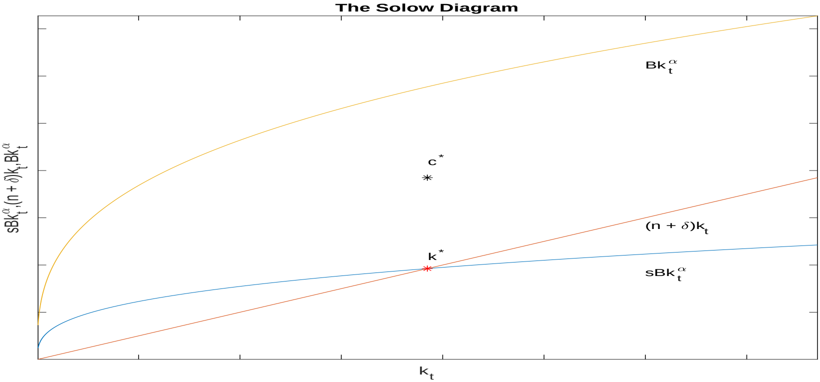
Interpretation: Additions to capital per worker come from investment (savings per worker), $sy_t = sBk_t^{\alpha}$, minus what is needed to compensate for depreciation and thinning out caused by population growth, $(n+\delta)k_t$.

Convergence to Steady State



- If $k_t < k^*$, investment exceeds depreciation, so that the capital stock will be expected to grow over time.
- If $k_t > k^*$, then depreciation and population growth exceed investment, and the capital stock will decline over time.

Golden Rule



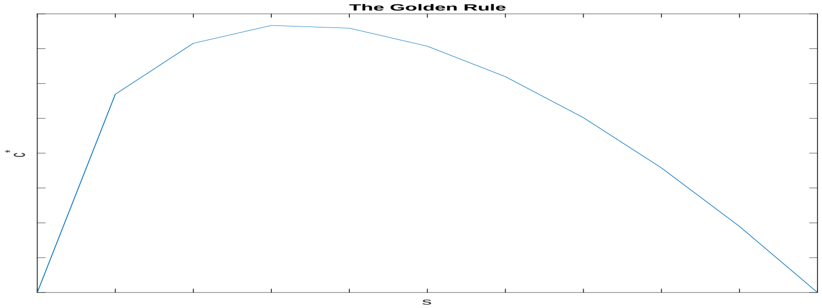
- c_t is given by the vertical distance between the plots of y_t and s_t, given a value of k_t.
- steady state occurs where the plot of sBk_t^α crosses the line $(n+\delta)k_t$.

Golden Rule

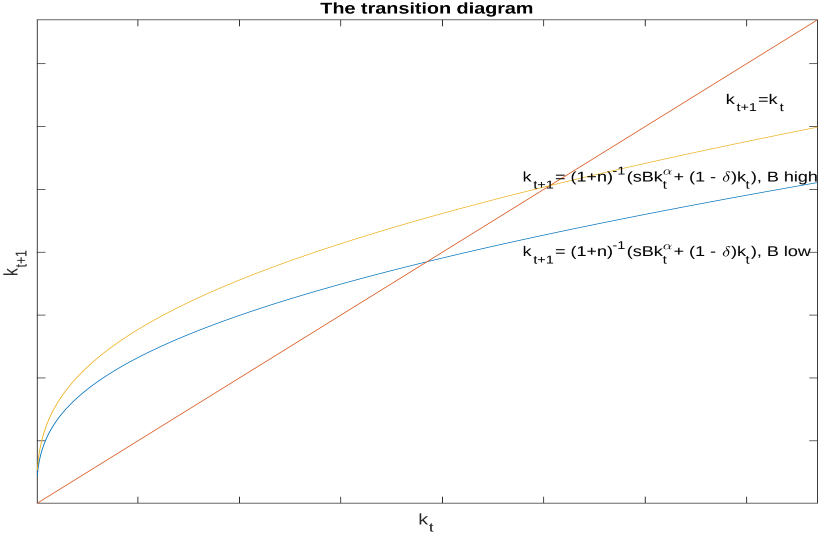
- the vertical distance is maximized when the slope of y_t is equal to the slope of the $(n+\delta)k_t$ plot, which means:

$$B\alpha(k_t)^{\alpha-1} = (n+\delta),$$

- implying that $k^{gr} = k^*$ if $s = \alpha$.
- The Golden rule s is the savings rate which generates a k^{gr} where the marginal product of capital equals $(n+\delta)$.



Permanent Increase in Technology



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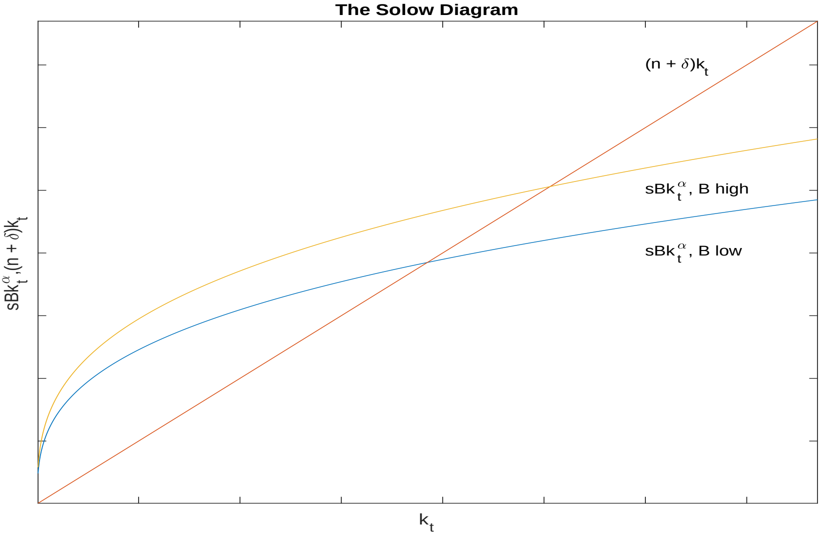


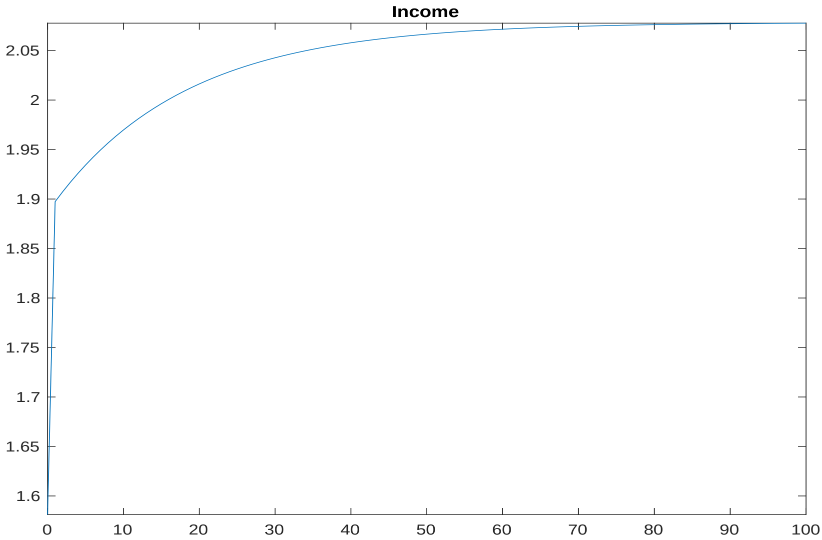
Permanent Increase in Technology

- k^*, y^*, c^* increase if B increases.
- Intuition: When the technology parameter B shifts upwards, production per worker and hence savings per worker increase, implying that now more capital is accumulated than needed to compensate for depreciation and population growth. As capital per worker increases, the marginal product of capital decreases until the steady state, where savings per worker just cover the now increased 'replacement investment' needed to compensate for depreciation and population growth.

Permanent Increase in Technology

- capital per worker, k^*, increases by 31.45%
- income per worker, y^*, increases by 31.45%
- consumption per worker, c^*, increases by 31.45%

Permanent Increase in Technology



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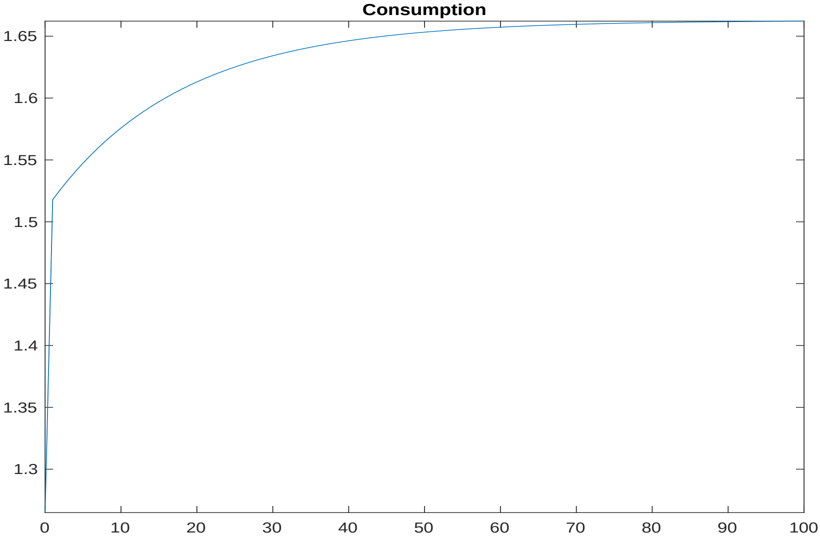
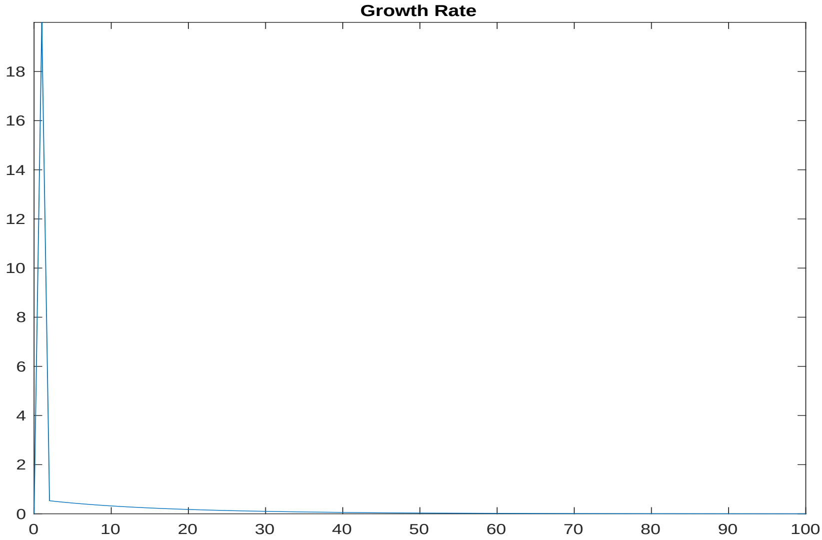
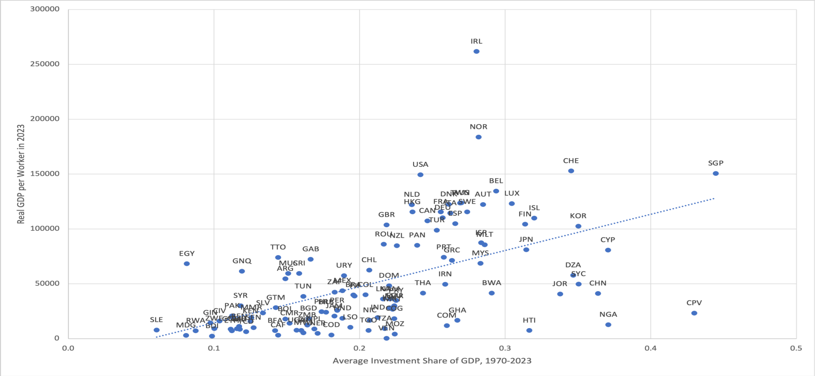


Permanent Increase in Technology

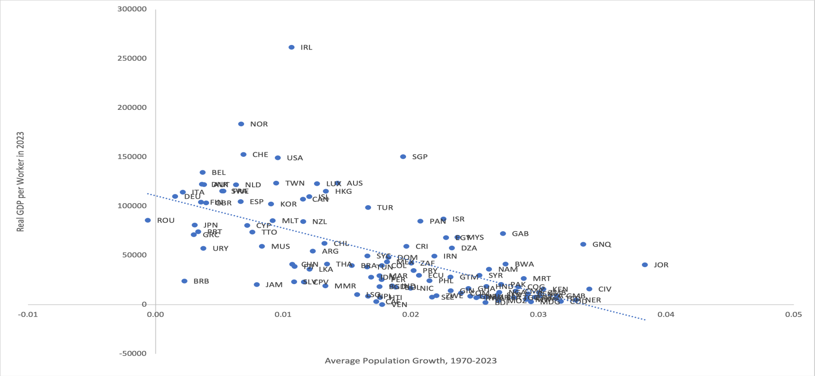


Conditional Convergence



Savings and investment accumulate as capital, and capital is productive, leading to more output. Hence, countries with higher savings rates are expected to have higher GDP per worker.

Conditional Convergence



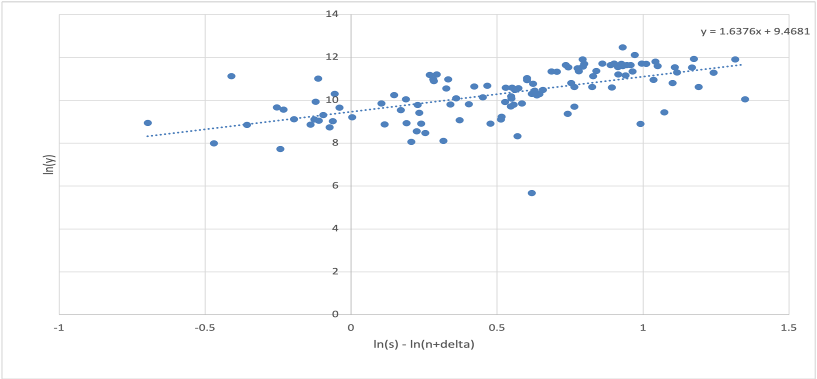
Higher population growth means higher growth of the labor force and thus a larger group of people who will share the capital accumulated in the past. Other things equal, this pulls GDP per worker down.

Conditional Convergence

The Solow model predicts that there should be a linear relationship between $\ln(y^*)$ and $\ln(s) - \ln(n+\delta)$, and that the slope should be $\frac{\alpha}{1-\alpha}$, which is equal to 0.5 if $\alpha = 1/3$:

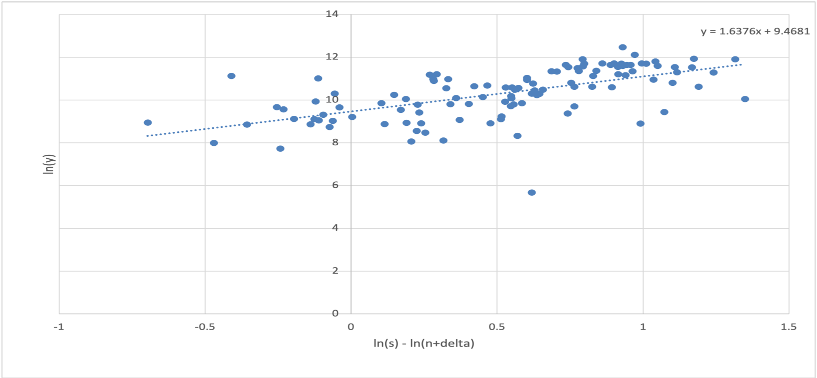
$$\ln(y^*) = \frac{1}{1-\alpha}\ln(B) + \frac{\alpha}{1-\alpha}\ln(s) - \frac{\alpha}{1-\alpha}\ln(n+\delta).$$

Conditional Convergence



The slope estimate is significantly positive, and the regression line seems to fit quite well, confirming the direction in the influence of $\ln(s) - \ln(n + \delta)$ on $ln(y)$ predicted by the theory as well as the linear relationship.

Conditional Convergence



The slope estimate is significantly larger than $1/2$, and we can conclude that in the real world the structural parameters s and n seem to have a stronger influence on y than predicted by the theory.